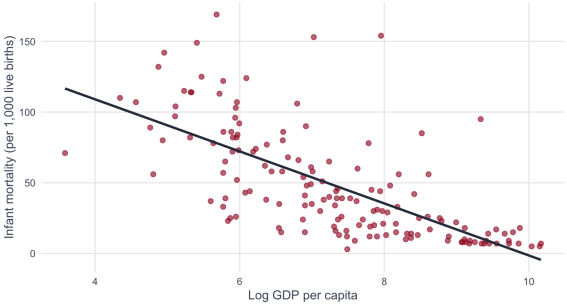
OLS: Estimation, Properties, and Inference

The sample analog of the BLP — and what we can learn from it

Gov 2001 · Scott Cunningham · Spring 2026

The question: why do some nations have ten times the infant mortality of others?



UN data, 154 countries (late 1990s)

- Range: 3–169 deaths per 1,000 live births
- Some countries 50× higher than others
- Why?

Model:

$$IM_i = \beta_0 + \beta_1 \ln(\text{GDP}_i) + \beta_2 TFR_i + \beta_3 Illit_i + \varepsilon_i$$

IM = infant mortality, TFR = fertility, $Illit$ = female illiteracy

The complete OLS output — we will walk through every piece of this today

OLS estimates ($n = 154$):

	$\hat{\beta}$	Naive SE	t
Intercept	53.34	11.29	4.72
ln(GDP)	−7.17	1.21	−5.95
Fertility	8.83	1.39	6.37
Illiteracy	0.52	0.09	6.05

$R^2 = 0.803 \quad F = 203.2$

BP test: $\chi^2(3) = 38.3$, $p < 0.001$

Naive vs. robust (HC3) SEs:

	Naive	Robust	Δ
Intercept	11.29	13.49	+19%
ln(GDP)	1.21	1.39	+15%
Fertility	1.39	1.56	+13%
Illiteracy	0.09	0.10	+18%

All SEs too small under naive formula.
Today we'll understand exactly why.

The BLP is a population object — OLS is how we estimate it from data

Population BLP (estimand)

$$\beta = \arg\min_b \mathbb{E}\left[(Y - X'b)^2\right]$$

Defined by population moments.
We can never observe it directly.

$\Longrightarrow$

OLS (estimator)

$$\hat{\beta} = \arg\min_b \frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i'b)^2$$

Same problem, sample moments replace population moments.

OLS is a **plug-in estimator**: replace $\mathbb{E}[\,\cdot\,]$ with $\frac{1}{n}\sum_i$ and solve the same FOC

OLS minimizes the sum of squared residuals — the FOC give the normal equations

Minimize over b_0, b_1 (bivariate for clarity):

$$\min_{b_0,\, b_1} \sum_{i=1}^{n} \left(Y_i - b_0 - b_1 X_i\right)^2$$

First-order conditions (set derivatives to zero):

$$\sum_{i=1}^{n} \hat{e}_i = 0 \qquad \sum_{i=1}^{n} X_i\, \hat{e}_i = 0 \qquad \text{where } \hat{e}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$$

$$\boxed{\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\widehat{\mathrm{Cov}}(X, Y)}{\widehat{\mathrm{Var}}(X)}}$$

The sample covariance-to-variance ratio — the exact sample analog of the population BLP slope.

In matrix form: $\hat{\beta} = (X'X)^{-1}X'Y$

Stack the data: $Y = X\beta + \varepsilon, \quad X$ is $n \times k, \quad Y$ is $n \times 1$

OLS solution:

$$\boxed{\hat{\beta} = (X'X)^{-1}X'Y}$$

Two requirements:

- $(X'X)$ must be invertible
- No perfect multicollinearity among columns of X

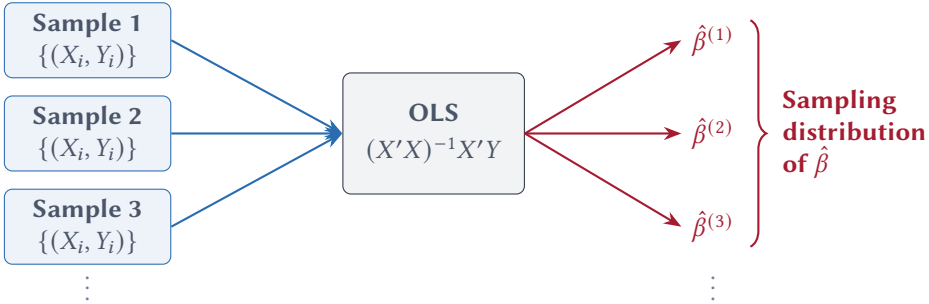
Residuals by construction:

$$\hat{e} = Y - X\hat{\beta}$$

$$X'\hat{e} = 0 \quad \text{(FOC, exactly)}$$

$\hat{\beta}$ is a function of the data (X, Y) — a random variable with its own sampling distribution.

OLS is a random variable — it has a sampling distribution



Each dataset gives a different $\hat{\beta}$. The distribution of $\hat{\beta}$ across datasets is what we analyze.

OLS is unbiased when $\mathbb{E}[\varepsilon \mid X] = 0$

Decompose $\hat{\beta}$:

$$\hat{\beta} = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon$$

Take expectations conditional on X:

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X' \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0 \text{ by assumption}} = \beta$$

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta \quad \text{— OLS is } \textbf{unbiased} \text{ under strict exogeneity}}$$

Finite-sample result: holds for any n, not just large n. The assumption $\mathbb{E}[\varepsilon_i \mid X_1, \ldots, X_n] = 0$ is stronger than what consistency needs.

Unbiasedness vs. consistency — two different properties, two different standards

Unbiased: $\mathbb{E}[\hat{\beta}] = \beta$

Centered on the truth *on average* across repeated samples

Finite-sample property. Holds for any n.

Consistent: $\hat{\beta} \xrightarrow{p} \beta$

Converges to the truth as $n \to \infty$

Asymptotic property. Requires LLN on moments.

- An estimator can be **biased but consistent** — e.g., MLE of σ^2 with n in denominator
- An estimator can be **unbiased but inconsistent** — e.g., Y_1 alone always targets μ but never converges
- OLS is **both** under standard assumptions

Unbiasedness is a stronger assumption. Consistency is weaker but what large samples guarantee.

OLS is consistent under a weaker condition — only $\mathbb{E}[X_i\varepsilon_i] = 0$ required

Why OLS is consistent:

$$\hat{\beta} - \beta = \left(\frac{1}{n}\sum_i X_i X_i'\right)^{-1} \frac{1}{n}\sum_i X_i \varepsilon_i$$

By LLN:

$$\frac{1}{n}\sum_i X_i X_i' \xrightarrow{p} \mathbb{E}[X_i X_i'] \equiv Q$$

$$\frac{1}{n}\sum_i X_i \varepsilon_i \xrightarrow{p} \mathbb{E}[X_i \varepsilon_i] = 0$$

By Slutsky:

$$\hat{\beta} - \beta \xrightarrow{p} Q^{-1} \cdot 0 = 0$$

$$\therefore \quad \hat{\beta} \xrightarrow{p} \beta$$

Consistency requires only $\mathbb{E}[X_i\varepsilon_i] = 0$ — a moment condition, not $\mathbb{E}[\varepsilon \mid X] = 0$

Unbiasedness needs the full $\mathbb{E}[\varepsilon \mid X] = 0$ (strict exogeneity). Consistency needs only the weaker version.

CLT gives normality to sample means — OLS is not a sample mean

CLT applies directly to simple averages:

$$\sqrt{n}\,(\bar{X} - \mu) \xrightarrow{d} N(0,\ \mathrm{Var}(X))$$

OLS is not a simple average:

$$\hat{\beta} = (X'X)^{-1}X'Y$$

- A ratio of matrix sums — nonlinear in the data
- CLT does not apply *directly* to $\hat{\beta}$
- Saying OLS is asymptotically normal is asserting something nontrivial

"Isn't everything normal under CLT?" — Only simple averages get normality for free. OLS requires showing it reduces to a scaled average. That's the content of the theorem.

OLS is approximately a scaled sample mean — that's why CLT applies

Decompose the estimation error:

$$\hat{\beta} - \beta = \underbrace{\left(\frac{X'X}{n}\right)^{-1}}_{\xrightarrow{p} Q^{-1} \text{ (LLN)}} \cdot \underbrace{\frac{X'\varepsilon}{n}}_{\text{sample mean of } X_i\varepsilon_i}$$

CLT on the numerator:

$$\sqrt{n} \cdot \frac{1}{n} \sum_i X_i \varepsilon_i \xrightarrow{d} N(0,\ \Sigma) \qquad \Sigma = \mathbb{E}[X_i X_i' \varepsilon_i^2]$$

Slutsky handles the denominator: multiply by $Q^{-1} \xrightarrow{p} Q^{-1}$

$$\boxed{\sqrt{n}\,(\hat{\beta} - \beta) \xrightarrow{d} N\big(0,\ Q^{-1}\Sigma Q^{-1}\big)}$$

CLT does the heavy lifting — but on $X_i\varepsilon_i$, not on Y_i directly. LLN stabilizes the denominator. Slutsky combines them.

Asymptotic normality is the foundation for every piece of inference we do

What $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$ gives us:

For large n:

$$\hat{\beta} \approx N\left(\beta, \frac{V}{n}\right)$$

We know the shape of the sampling distribution — even though β is unknown.

What we still need:

An estimate $\hat{V}$ of V

$\hat{V}$ determines our standard errors — and getting it wrong breaks all inference.

The question "what is V?" is exactly the homoskedasticity vs. heteroskedasticity question — coming up next.

Gauss-Markov: OLS is BLUE under four assumptions

The four classical assumptions:

1. **Linearity:** $Y_i = X_i'\beta + \varepsilon_i$
2. **Full rank:** $(X'X)$ invertible — no perfect multicollinearity
3. **Strict exogeneity:** $\mathbb{E}[\varepsilon_i \mid X_1, \ldots, X_n] = 0$
4. **Homoskedasticity:** $\text{Var}(\varepsilon_i \mid X) = \sigma^2$ for all i

Gauss-Markov Theorem: Under (1)–(4), OLS is **BLUE** — Best Linear Unbiased Estimator

Among all estimators of the form $\tilde{\beta} = AY$ that satisfy $\mathbb{E}[\tilde{\beta}] = \beta$, OLS achieves the smallest variance for every coefficient.

BLUE: what each word means

B	L	U	E
Best	**Linear**	**Unbiased**	**Estimator**
Minimum variance	$\tilde{\beta} = AY$ a linear function of the data	$\mathbb{E}[\tilde{\beta}] = \beta$ centered on the truth	$\hat{\beta}_k$, one component at a time

- "Best among linear unbiased" — not best among *all* estimators
- Finite-sample result: no asymptotic argument needed
- **Requires homoskedasticity (assumption 4)** — this is the critical one
- Under heteroskedasticity: OLS is still unbiased, but **no longer best**

WLS (weighted least squares) is BLUE under heteroskedasticity. But we usually just fix the SEs.

What you get — and what you need — under each assumption

Assumptions held	OLS property	What breaks without it
$\mathbb{E}[\varepsilon \mid X] = 0$	Unbiased	Point estimates are off
+ Homoskedasticity	BLUE (Gauss-Markov)	OLS is inefficient; WLS beats it
+ Large n (LLN, CLT)	Consistent + asymp. normal	No sampling distribution
Heterosk. only	Unbiased, consistent, normal	**Standard errors are wrong**

Heteroskedasticity does not bias $\hat{\beta}$ — it corrupts our estimate of $\text{Var}(\hat{\beta})$

This is why heteroskedasticity is fundamentally an inference problem, not an estimation problem.

The variance of $\hat{\beta}$ is the pivot of all inference

From asymptotic normality: $\hat{\beta}_k \approx N\Big(\beta_k,\ \mathrm{Var}(\hat{\beta}_k)\Big)$

Every inferential object depends on $\mathrm{Var}(\hat{\beta}_k)$**:**

Standard error	**t-statistic**	**Confidence interval**
$\mathrm{SE}(\hat{\beta}_k) = \sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_k)}$	$t = \dfrac{\hat{\beta}_k - \beta_k^0}{\mathrm{SE}(\hat{\beta}_k)}$	$\hat{\beta}_k \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_k)$

Wrong $\widehat{\mathrm{Var}}(\hat{\beta}_k) \implies$ wrong SE $\implies$ wrong t $\implies$ wrong p-values and CIs

Getting the variance formula right is not a detail. It is the whole inference problem.

Homoskedasticity: one number summarizes the error variance everywhere

Assumption: $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ (constant, does not depend on X_i)

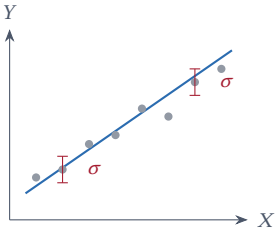
Population variance of $\hat{\beta}$:

$$\text{Var}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

Estimated:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$$

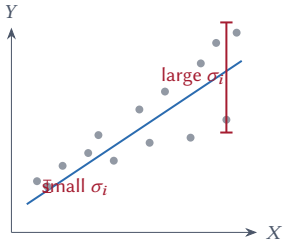
$$\hat{\sigma}^2 = \frac{1}{n-k} \sum_{i=1}^{n} \hat{e}_i^2$$



One number $\hat{\sigma}^2$ scales the whole variance matrix. Clean — but only valid under homoskedasticity.

Heteroskedasticity: the error variance changes with X — the homoskedastic formula breaks

Fan-shaped residuals:



$\text{Var}(\varepsilon_i \mid X_i) = \sigma_i^2$ **varies with** X_i

The homoskedastic formula uses $\hat{\sigma}^2$ **— a single average:**

$$\widehat{\text{Var}}_{hom}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$$

- Over-smooths the true variance structure
- Wrong in every part of the X range
- SEs too small where variance is large
- SEs too large where variance is small

Result: t-statistics, p-values, and CIs are all wrong.

Three statisticians solved the problem — White brought it to econometrics

Eicker (1963)

German statistician

Proved the sandwich estimator is *consistent* for the asymptotic variance under heteroskedasticity

Ann. Math. Stat. 1963

Huber (1967)

Swiss statistician

Extended the theory to M-estimators broadly — robust statistics as a general framework

Berkeley Symp. 1967

White (1980)

UC San Diego economist

Synthesized Eicker and Huber for econometrics; added the White test for heteroskedasticity

Econometrica 1980

The estimator is called **Eicker-Huber-White (EHW)** or **HC** (heteroskedasticity-consistent) standard errors

White's paper made the idea practical enough that economists actually started using it. Before 1980, nearly everyone reported naive SEs by default.

The sandwich estimator lets each residual speak for itself

Eicker-Huber-White (robust) variance estimator:

$$\widehat{\text{Var}}_{robust}(\hat{\beta}) = \underbrace{(X'X)^{-1}}_{\text{bread}} \underbrace{\left(\sum_{i=1}^{n} \hat{e}_i^2 \, X_i X_i'\right)}_{\text{meat}} \underbrace{(X'X)^{-1}}_{\text{bread}}$$

Why it works:

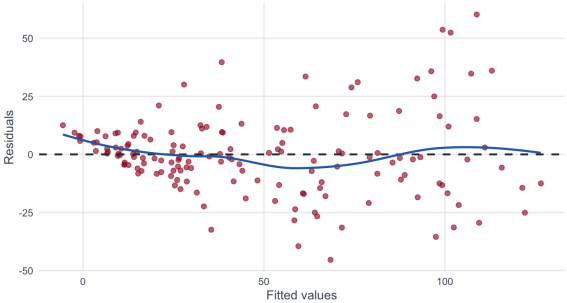
- $\hat{e}_i^2$ estimates σ_i^2 at each observation
- Weights $X_i X_i'$ by the local variance
- No assumption on the shape of σ_i^2

In practice:

- `lm_robust()` in R (`estimatr`)
- `vce(robust)` in Stata
- Use by default — the cost of not using it can be severe

Angrist & Pischke: "always report heteroskedasticity-robust standard errors." Homoskedasticity is an assumption; robust SEs are not.

Back to UN98: the residuals reveal a clear fan shape



What we see:

- Low fitted values: residuals spread widely
- High fitted values: residuals compress
- Classic fan shape — variance falls as predicted mortality rises

Breusch-Pagan test:

$$\chi^2(3) = 38.3, \quad p < 0.001$$

The naive formula uses one $\hat{\sigma}^2$ for all 154 countries. It over-weights low-variance (rich) countries and under-weights high-variance (poor)

Back to UN98: naive SEs are too small — robust SEs are 13–19% larger

	Naive SE	Robust SE	Change
Intercept	11.29	13.49	+19%
ln(GDP)	1.21	1.39	+15%
Fertility	1.39	1.56	+13%
Illiteracy	0.09	0.10	+18%

$\hat{\sigma}^2$ is too small because it averages over low-variance (high-income) and high-variance (low-income) countries together.

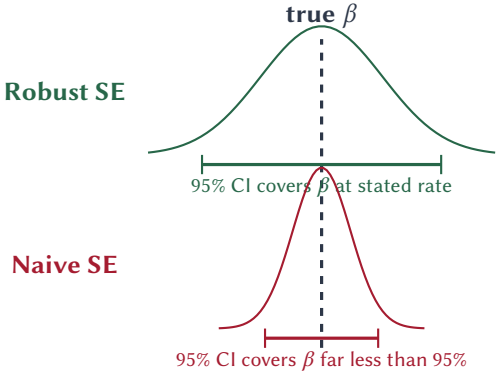
95% CI for $\hat{\beta}_{\ln(\mathbf{GDP})}$:

Naive: $[-9.53,\ -4.81]$
Robust: $[-9.88,\ -4.45]$

Robust CI is **15% wider** (5.43 vs. 4.72 in width)

The naive formula tells us we know $\beta_{\ln(GDP)}$ more precisely than we actually do

What heteroskedasticity does to inference — a visual



The estimator $\hat{\beta}$ is in the same place — unbiasedness is intact. But the spread we *report* for it is wrong. That is the entire problem.

t-statistic and confidence interval for a single coefficient $\hat{\beta}_k$

From asymptotic normality:

$$\frac{\hat{\beta}_k - \beta_k}{\mathrm{SE}(\hat{\beta}_k)} \xrightarrow{d} N(0,1)$$

Hypothesis test ($H_0 : \beta_k = 0$):

$$t = \frac{\hat{\beta}_k}{\mathrm{SE}(\hat{\beta}_k)}$$

Reject H_0 at 5% if $|t| > 1.96$

p-value $= 2\mathbb{P}(Z > |t|)$, $Z \sim N(0,1)$

95% confidence interval:

$$\hat{\beta}_k \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_k)$$

Covers the true β_k in 95% of repeated samples

Use t_{n-k} critical values for small n

Same logic as the inference lectures — now $\mathrm{SE}(\hat{\beta}_k) = \sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_k)/n}$

Robust or not robust: the formula is the same. What changes is how we estimate $\mathrm{Var}(\hat{\beta}_k)$.

F-test: testing joint hypotheses on multiple coefficients

Why not just run multiple t-tests?

- Testing $\beta_1 = 0$ and $\beta_2 = 0$ separately inflates the Type I error rate
- A joint test holds the false rejection rate at α for the *joint* null

Restricted vs. unrestricted models:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n-k)} \sim F_{q,\,n-k} \text{ under } H_0$$

- SSR_R: sum of squared residuals imposing H_0
- SSR_U: sum of squared residuals of the full model
- q: number of restrictions
- Intuition: how much does imposing H_0 hurt the fit?

Special case: $q = 1 \Rightarrow F = t^2$. The F-test is the generalization of the t-test to multiple restrictions.

F-test example: are all slope coefficients jointly zero?

Null: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_{k-1} = 0$

Restricted model (intercept only):

$$Y_i = \beta_0 + \varepsilon_i$$

$$SSR_R = \sum (Y_i - \bar{Y})^2 = TSS$$

Unrestricted model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \varepsilon_i$$

$$SSR_U = \sum \hat{e}_i^2$$

$$F = \frac{(TSS - SSR_U)/(k-1)}{SSR_U/(n-k)} = \frac{R^2/(k-1)}{(1-R^2)/(n-k)}$$

When F is large, the regressors together explain enough variance to reject H_0

This is the "overall F-statistic" reported by every regression package. It answers: does the model explain anything at all?

R^2 is the fraction of Y-variance explained by the regression

Variance decomposition:

$$\underbrace{\sum_i (Y_i - \bar{Y})^2}_{TSS} = \underbrace{\sum_i (\hat{Y}_i - \bar{Y})^2}_{ESS} + \underbrace{\sum_i \hat{e}_i^2}_{SSR}$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS} \in [0, 1]$$

- $R^2 = 0$: regression explains nothing (flat line through $\bar{Y}$)
- $R^2 = 1$: perfect fit, all points on the line
- Bivariate OLS: $R^2 = \widehat{\text{Corr}}(X, Y)^2$

Adding *any* regressor — even noise — can only increase R^2 or leave it flat. Never decreases.

Adjusted R^2 penalizes complexity — $\bar{R}^2$ can fall when you add irrelevant variables

Adjusted R^2:

$$\bar{R}^2 = 1 - \frac{SSR/(n-k)}{TSS/(n-1)} = 1 - (1 - R^2)\frac{n-1}{n-k}$$

What it does:

- Divides by degrees of freedom, not raw sums
- Penalizes adding regressors that don't reduce SSR enough
- $\bar{R}^2$ *can* decrease — signals the new variable wasn't worth it

When $\bar{R}^2$ falls:

$$\Delta\bar{R}^2 < 0 \iff F_{\text{new var}} < 1$$

The new variable's F-statistic is less than 1 — it actively hurts the adjusted fit.

$\bar{R}^2$ is a better model-comparison tool than R^2 — but neither is a causal object.

What R^2 tells you — and what it doesn't

R^2 does tell you:

- How well the line fits this sample
- How much variance is explained by the included regressors
- Whether two models fit differently (via $\bar{R}^2$)

R^2 does NOT tell you:

- Whether $\hat{\beta}$ is unbiased or consistent
- Whether the causal interpretation is valid
- Whether the standard errors are correct
- Whether you should add more variables

Low R^2 with correct specification beats high R^2 with the wrong model

Causal inference cares about the specification, not the fit. $R^2 = 0.04$ with a valid instrument beats $R^2 = 0.90$ with endogenous regressors.

Today: OLS as an estimator with a sampling distribution — next: robust SE applications

The arc of today:

- **OLS** is the sample plug-in for the BLP: $\hat{\beta} = (X'X)^{-1}X'Y$
- **Unbiased** under strict exogeneity — finite-sample result
- **Consistent** under the weaker moment condition $\mathbb{E}[X_i\varepsilon_i] = 0$
- **Asymptotically normal** because $\frac{X'\varepsilon}{n}$ is a sample mean — CLT + Slutsky
- **BLUE** (Gauss-Markov) under homoskedasticity — loses "B" under heteroskedasticity
- **Heteroskedasticity** is an inference problem: use the sandwich estimator
- **Inference**: t-stats and CIs for single $\hat{\beta}_k$; F-tests for joint hypotheses
- R^2 measures fit, not correctness — $\bar{R}^2$ penalizes complexity